Lab 3: Reading the fingerprints of ecological count data

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```
library(tidyverse)
library(fitdistrplus)
library(MASS)      # glm.nb
library(glmmTMB)   # mixed models, zero inflation
library(DHARMa)    # diagnostics
library(readxl)

set.seed(123)
theme_set(theme_minimal(base_size = 12))
```

Learning objectives

By the end of this class, you should be able to:

1. Describe what a single row represents in a dataset and how it was collected

2.Identify distributional fingerprints (zeros, skew, mean–variance relationships)

3.Decide whether Poisson, Negative Binomial, zero-inflated, or hierarchical models are plausible

4.Explain how sampling conditions affect abundance indices

5.Recognize when extreme-value covariates can mislead ecological inference

6.Translate data structure → shortlist of analysis methods

##PART I — Simulated tick count data (30 minutes) 1. Data-collection story

We sampled adult ticks using drag cloths at three sites, repeatedly through the season and across years.

At each visit, we recorded:

-temp_capture, rh_capture: microclimate at the time of sampling

-tmax_annual, rhmin_annual: remote-sensed annual extremes

-tick_count: number of adult ticks collected

Key idea: Observed tick counts = true population × questing probability × sampling conditions

Microclimate controls questing activity, so it affects confidence in the count as an index of abundance. 2. Simulate repeated-measures tick data

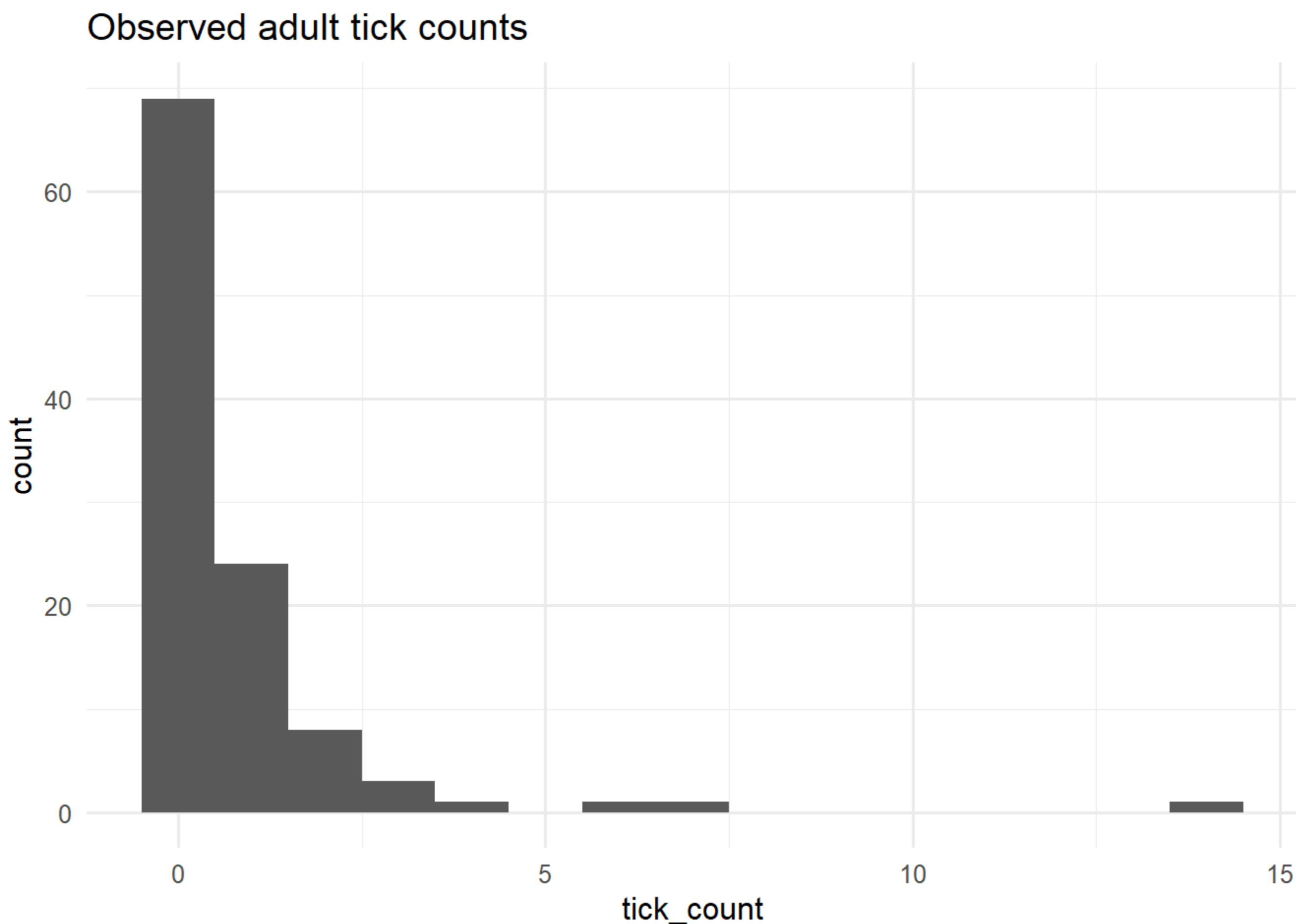
```
sites <- c("A","B","C")
years <- c(2024, 2025)

tick_dat <- expand_grid(
  site = sites,
  year = years,
  visit = 1:18
) %>%
  mutate(
    doy = round(scales::rescale(visit, to = c(130, 270))),

    # Annual extremes (constant within site-year)
    tmax_annual = rnorm(n(), 33, 2) + if_else(site=="C", 1.5, 0),
    rhmin_annual = pmax(15, rnorm(n(), 30, 7) - if_else(site=="B", 4, 0))
  ) %>%
  group_by(site, year) %>%
  mutate(
    tmax_annual = first(tmax_annual),
    rhmin_annual = first(rhmin_annual)
  ) %>%
  ungroup() %>%
  mutate(
    temp_capture = rnorm(n(), 22, 5),
    rh_capture   = pmax(35, rnorm(n(), 70, 10))
  )
```

3. Generate tick counts with a questing process

```
tick_dat <- tick_dat %>%
  mutate(
    # True underlying abundance (site + year signal)
    log_abundance = 1.2 +
      if_else(site=="A", 0.3, 0) +
      if_else(site=="C", -0.4, 0) +
      if_else(year==2025, 0.2, 0),

    # Questing probability depends on capture microclimate
    questing = plogis(
      -3 +
      0.18 * temp_capture -
      0.04 * rh_capture
    ),

    mu = exp(log_abundance) * questing,

    tick_count = rnbinom(n(), mu = mu, size = 1.1)
  )
```

4. First fingerprint: what do the counts look like?

```
tick_dat %>%
  ggplot(aes(tick_count)) +
  geom_histogram(binwidth = 1) +
  labs(title = "Observed adult tick counts")
```




```
mean(tick_dat$tick_count)
```

```
## [1] 0.7407407
```

```
var(tick_dat$tick_count)
```

```
## [1] 2.997577
```

```
mean(tick_dat$tick_count == 0)
```

```
## [1] 0.6388889
```

Discussion time!

Does our mean roughly equal our variance? For count data we'll often, as ecologists, think about the Poisson distribution. The Poisson assumes that variance and our mean are roughly equivalent. Is that the case here?

What is greater?

Roughly what proportion of our data is zeros? Generally speaking >50% zeros implies zero-inflation, BUT this does not mean we need to use zero-inflation techniques yet! What kind of zeros we have is important for how we deal with them.

Are all of our data points in this dataset independent spatially and temporally? No, we resampled our sites, so we cannot treat all of these data points like they came from the same population.

Fingerprint points toward: Negative Binomial ± zero inflation ± random effects

5. Distribution fitting (quick check)

```
fit_pois <- fitdist(tick_dat$tick_count, "pois")
fit_nb   <- fitdist(tick_dat$tick_count, "nbinom")

fit_pois$aic
```

```
## [1] 318.805
```

```
fit_nb$aic
```

```
## [1] 252.7714
```

Why Poisson fails here

Poisson assumes:

-Events occur independently

-Constant rate

-Mean = variance

-No unmodeled heterogeneity

But our tick data have:

-Repeated measures

-Seasonal phenology

-Microclimate-driven availability

-Unobserved site- and visit-level variation

So the Poisson model is being asked to do something ecologically impossible! This makes our job more difficult but also really cool!

6. Models with microclimate at capture vs annual extremes

Model A: Capture conditions (mechanistic)

```
m_micro <- glmmTMB(
  tick_count ~ temp_capture + rh_capture + (1|site),
  family = nbinom2(),
  data = tick_dat
)
summary(m_micro)
```

```
##  Family: nbinom2  ( log )
## Formula:          tick_count ~ temp_capture + rh_capture + (1 | site)
## Data: tick_dat
##
##      AIC       BIC    logLik -2*log(L)  df.resid
##    230.2     243.6    -110.1     220.2       103
##
## Random effects:
##
## Conditional model:
##  Groups Name        Variance Std.Dev.
##  site   (Intercept) 0.1349   0.3673
## Number of obs: 108, groups:  site, 3
##
## Dispersion parameter for nbinom2 family (): 1.06
##
## Conditional model:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.10000    1.23732  -1.697  0.08966 .
## temp_capture  0.18894    0.03657   5.166 2.39e-07 ***
## rh_capture   -0.04157    0.01612  -2.579  0.00992 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

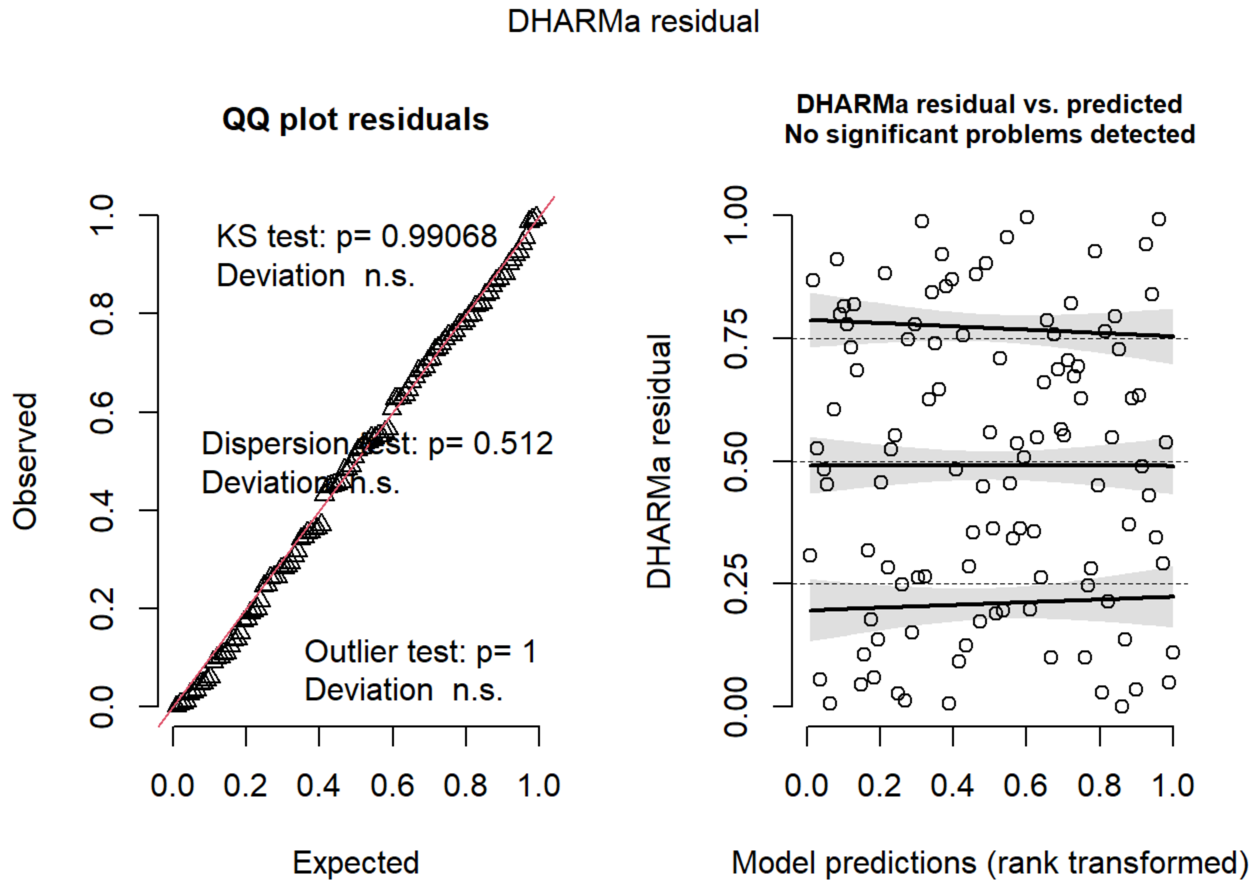
Model B: Annual extremes (ecological but indirect)

```
m_extreme <- glmmTMB(
  tick_count ~ tmax_annual + rhmin_annual + (1|site),
  family = nbinom2(),
  data = tick_dat
)
summary(m_extreme)
```

```
##  Family: nbinom2  ( log )
## Formula:          tick_count ~ tmax_annual + rhmin_annual + (1 | site)
## Data: tick_dat
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     254.3     267.7    -122.2     244.3       103
##
## Random effects:
##
## Conditional model:
##  Groups Name        Variance  Std.Dev.
##  site   (Intercept) 1.507e-09 3.882e-05
## Number of obs: 108, groups:  site, 3
##
## Dispersion parameter for nbinom2 family (): 0.477
##
## Conditional model:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.64589    3.82778   0.169   0.8660
## tmax_annual  -0.07205    0.11199  -0.643   0.5200
## rhmin_annual  0.06246    0.02985   2.093   0.0364 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

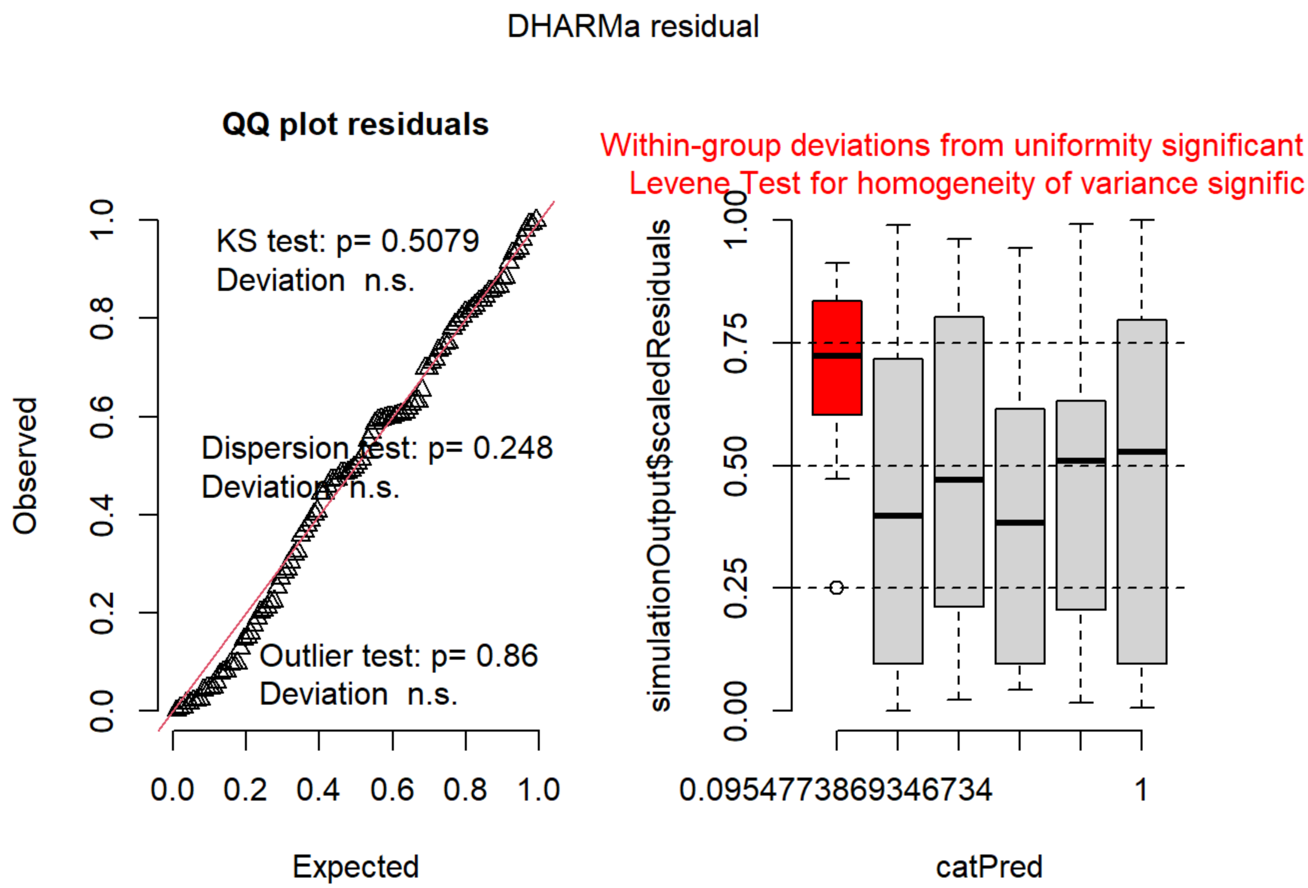
7. Diagnostics & interpretation

```
simulateResiduals(m_micro, plot = TRUE)
```



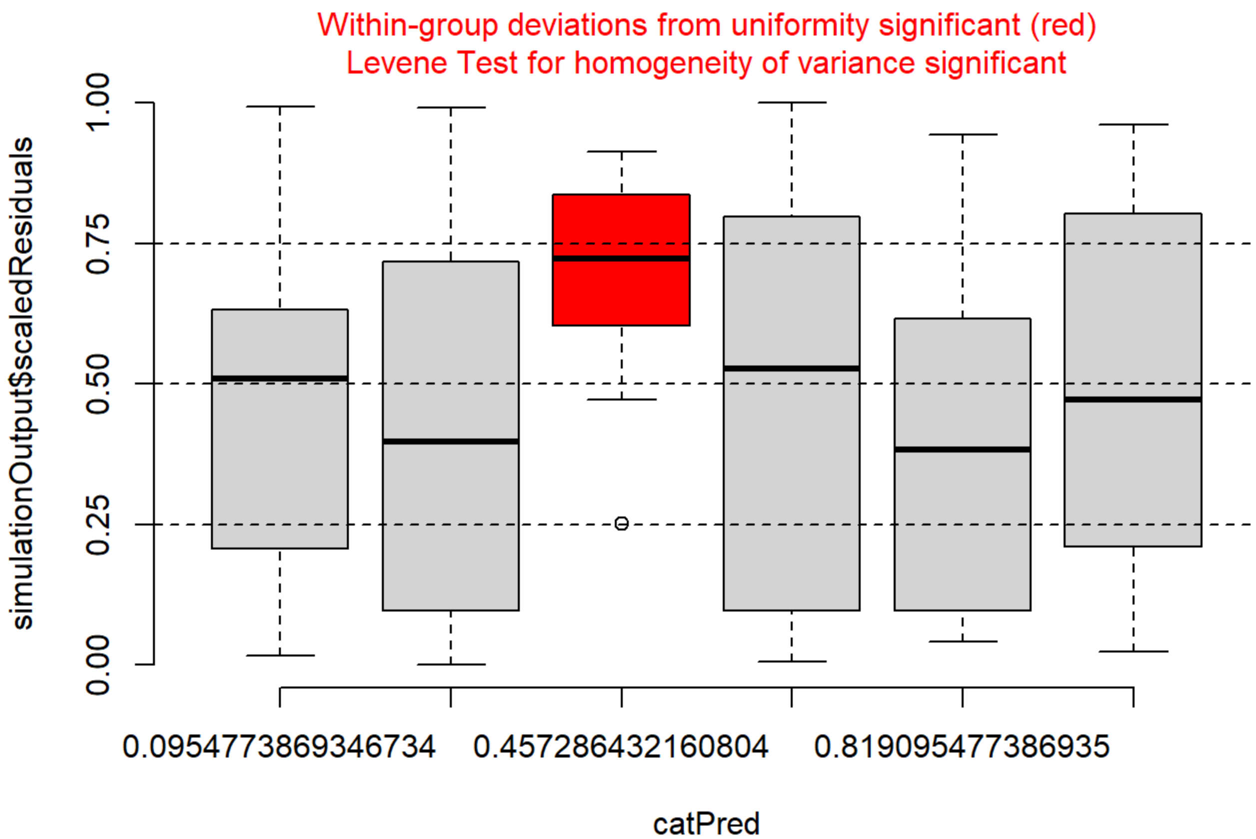

```
## Object of Class DHARMa with simulated residuals based on 250 simulations with refit = FALS
E . See ?DHARMa::simulateResiduals for help.
##
## Scaled residual values: 0.4837926 0.1781393 0.7789354 0.0546218 0.8195241 0.7575531 0.5648
41 0.1368309 0.8808549 0.01195136 0.8827028 0.3421742 0.6346702 0.1358446 0.9422971 0.1738816
0.3538355 0.6848726 0.7988134 0.9110042 ...
```

```
simulateResiduals(m_extreme, plot = TRUE)
```

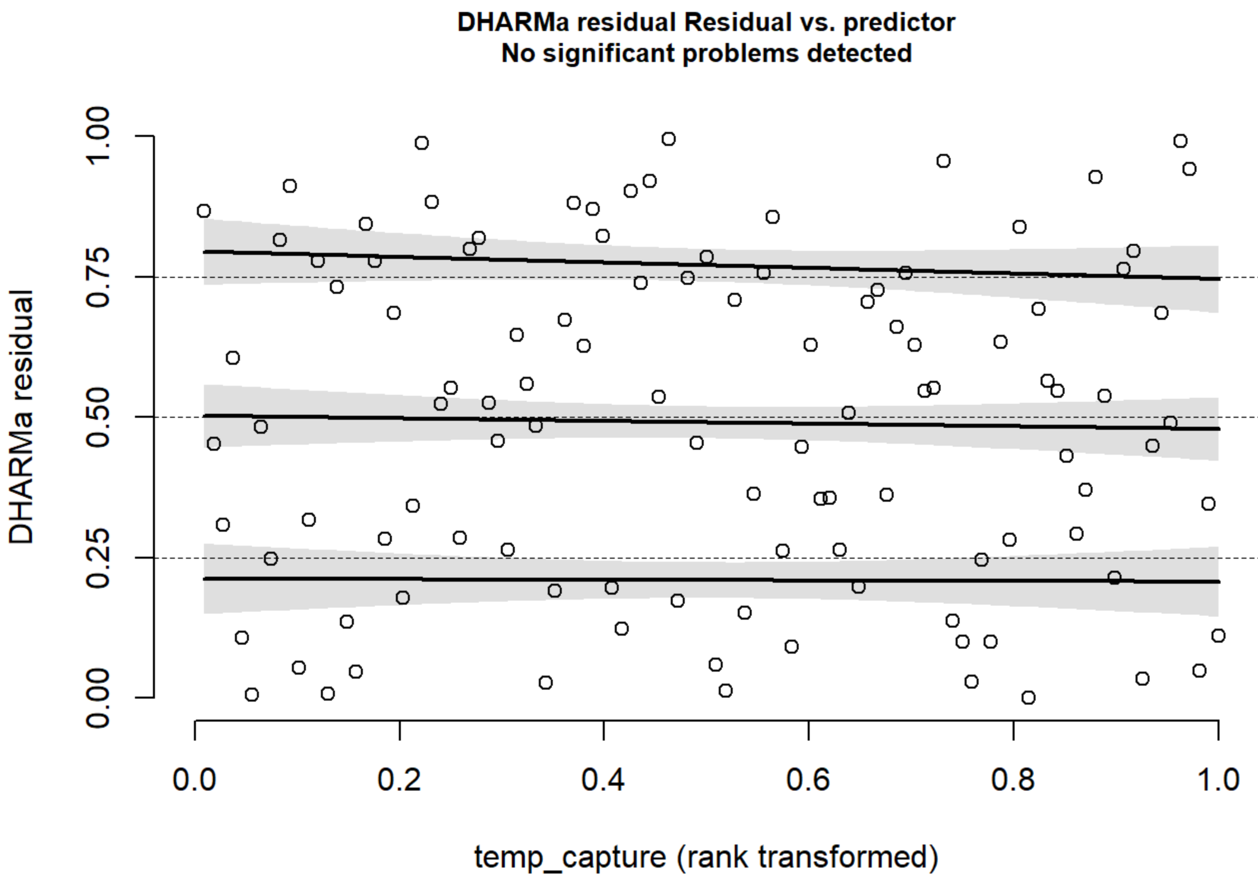



```
## Object of Class DHARMa with simulated residuals based on 250 simulations with refit = FALS
E . See ?DHARMa::simulateResiduals for help.
##
## Scaled residual values: 0.5673933 0.0164542 0.4445526 0.5510645 0.318215 0.7348601 0.63177
73 0.3030822 0.6980213 0.2060698 0.6042238 0.1505179 0.8271681 0.5051669 0.9929844 0.5128016
0.1581304 0.1267464 0.006383884 0.5980159 ...
```

```
#Compare this
plotResiduals(
  simulateResiduals(m_extreme),
  tick_dat$tmax_annual
)
```




```
#to this
plotResiduals(
  simulateResiduals(m_micro),
  tick_dat$temp_capture
)
```



Reflection time!

Microclimate explains when ticks are available to be sampled

Annual extremes push models toward boundary values

heteroskedasticity = across group variation (not great for our current model formulation)

Extremes may describe range limits, not local abundance processes

This is why MaxEnt-style and Max likelihood approaches often gravitate toward extremes that are not mechanistically tied to the sampling event.

Let's explore adding a seasonal curve using splines/a GAM approach!

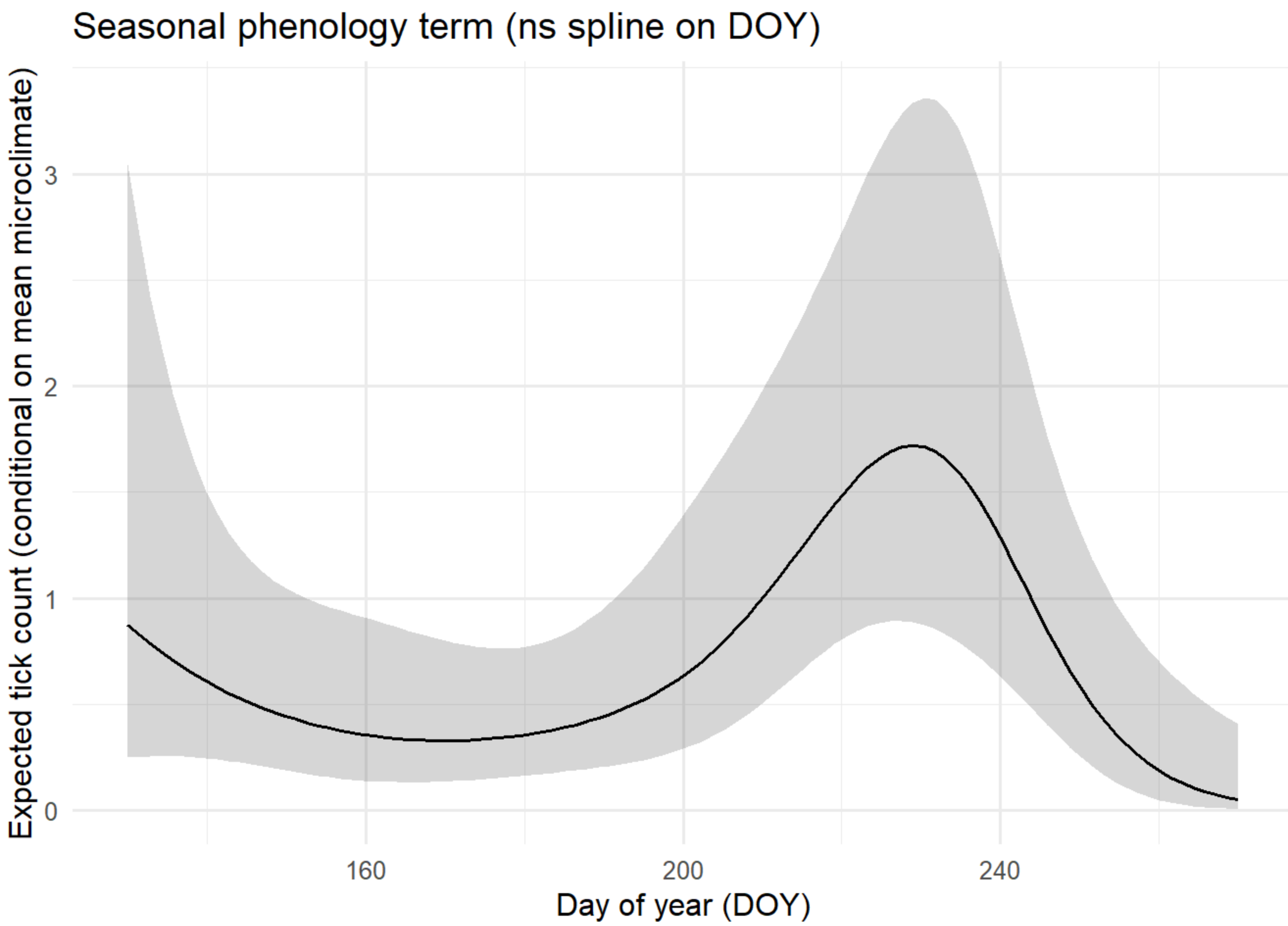
```
library(splines)  # ns()
m_micro_season <- glmmTMB(
  tick_count ~ ns(doy, df = 4) + temp_capture + rh_capture + (1|site),
  family = nbinom2(),
  data = tick_dat
)

summary(m_micro_season)
```

```
##  Family: nbinom2  ( log )
## Formula:
## tick_count ~ ns(doy, df = 4) + temp_capture + rh_capture + (1 |      site)
## Data: tick_dat
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     218.5     242.7    -100.3     200.5        99
##
## Random effects:
##
## Conditional model:
##  Groups Name        Variance Std.Dev.
##  site   (Intercept) 0.1162   0.3409
## Number of obs: 108, groups:  site, 3
##
## Dispersion parameter for nbinom2 family (): 1.91
##
## Conditional model:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -1.75704    1.16940  -1.503  0.13297
## ns(doy, df = 4)1 -0.60894    0.81752  -0.745  0.45635
## ns(doy, df = 4)2  2.08408    0.74620   2.793  0.00522 **
## ns(doy, df = 4)3 -2.44041    1.50510  -1.621  0.10493
## ns(doy, df = 4)4 -2.11301    1.16435  -1.815  0.06956 .
## temp_capture      0.17083    0.03423   4.990 6.03e-07 ***
## rh_capture       -0.03694    0.01543  -2.393  0.01669 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

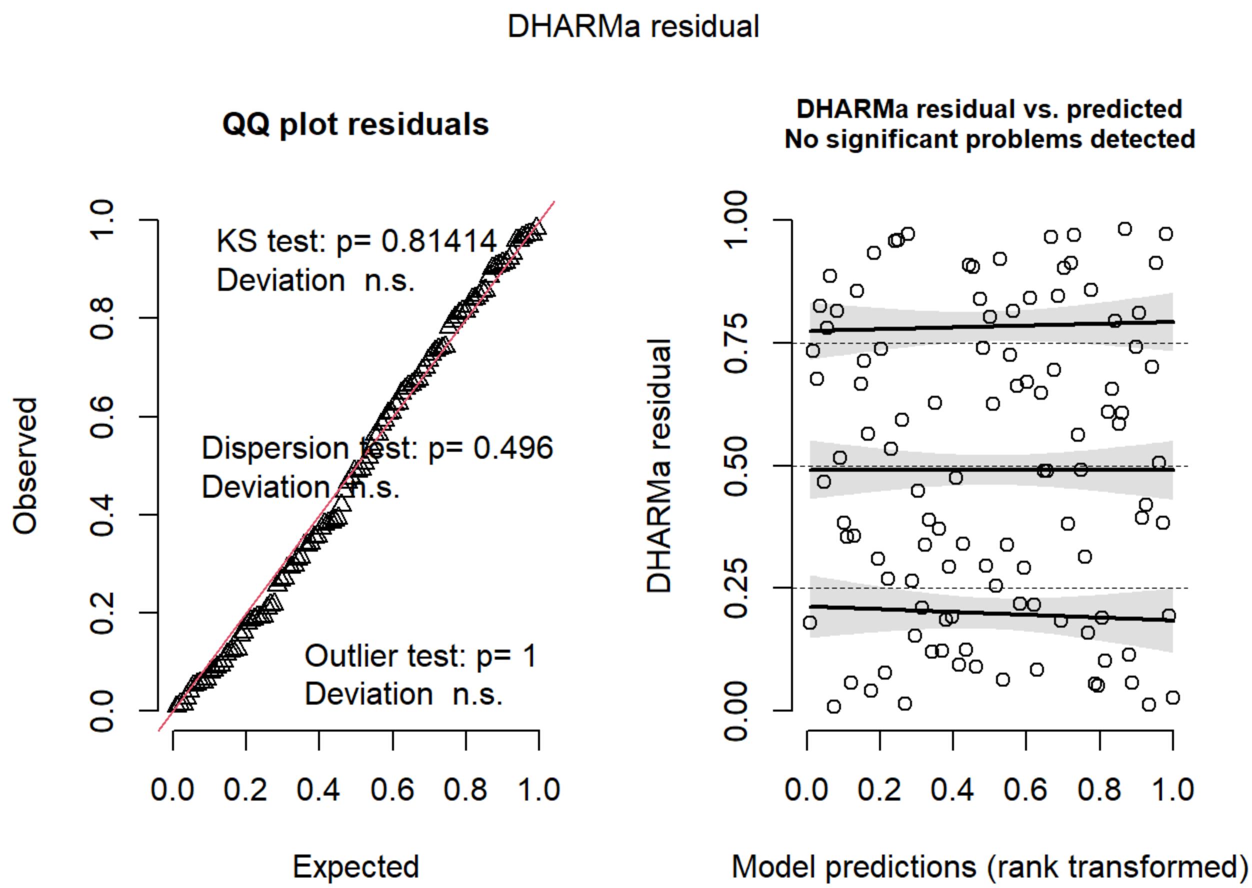
```
m_micro_noseason <- glmmTMB(
  tick_count ~ temp_capture + rh_capture + (1|site),
  family = nbinom2(),
  data = tick_dat
)

AIC(m_micro_noseason, m_micro_season)
```

```
##                  df      AIC
## m_micro_noseason  5 230.1970
## m_micro_season    9 218.5241
```

```
new_doy <- tibble(
  doy = seq(min(tick_dat$doy), max(tick_dat$doy), length.out = 100),
  temp_capture = mean(tick_dat$temp_capture),
  rh_capture   = mean(tick_dat$rh_capture),
  site = tick_dat$site[1]  # pick a reference site for prediction
)

pred <- predict(m_micro_season, newdata = new_doy, type = "link", se.fit = TRUE)

new_doy <- new_doy %>%
  mutate(
    eta = pred$fit,
    se  = pred$se.fit,
    mu  = exp(eta),
    lo  = exp(eta - 1.96 * se),
    hi  = exp(eta + 1.96 * se)
  )

ggplot(new_doy, aes(doy, mu)) +
  geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2) +
  geom_line() +
  labs(
    title = "Seasonal phenology term (ns spline on DOY)",
    y = "Expected tick count (conditional on mean microclimate)",
    x = "Day of year (DOY)"
  )
```




```
DHARMa::simulateResiduals(m_micro_season, plot = TRUE)
```




```
## Object of Class DHARMa with simulated residuals based on 250 simulations with refit = FALS
E . See ?DHARMa::simulateResiduals for help.
##
## Scaled residual values: 0.08306169 0.01399167 0.5644563 0.8144662 0.05806845 0.7403847 0.7
259862 0.4921931 0.9081383 0.3409698 0.905679 0.1905584 0.3840496 0.2917469 0.9132548 0.62752
07 0.3569091 0.7342977 0.2655035 0.9328318 ...
```

-DOY captures a latent seasonal process that influences the probability ticks are questing and thus detectable.

-Microclimate-at-capture captures instantaneous availability.

-The spline absorbs broad seasonal swings so that temp_capture and rh_capture aren’t forced to “explain” phenology.

```
#inside the normal gam package this is what a seasonal spline might look like
# install.packages("mgcv") if needed
library(mgcv)
```

```
## Loading required package: nlme
```

```
##
## Attaching package: 'nlme'
```

```
## The following object is masked from 'package:dplyr':
##
##     collapse
```

```
## This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.
```

```
tick_dat <- tick_dat %>%
  mutate(
    site = factor(site),
    doy  = as.numeric(doy)
  )

summary(tick_dat$doy)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     130     163     200     200     237     270
```

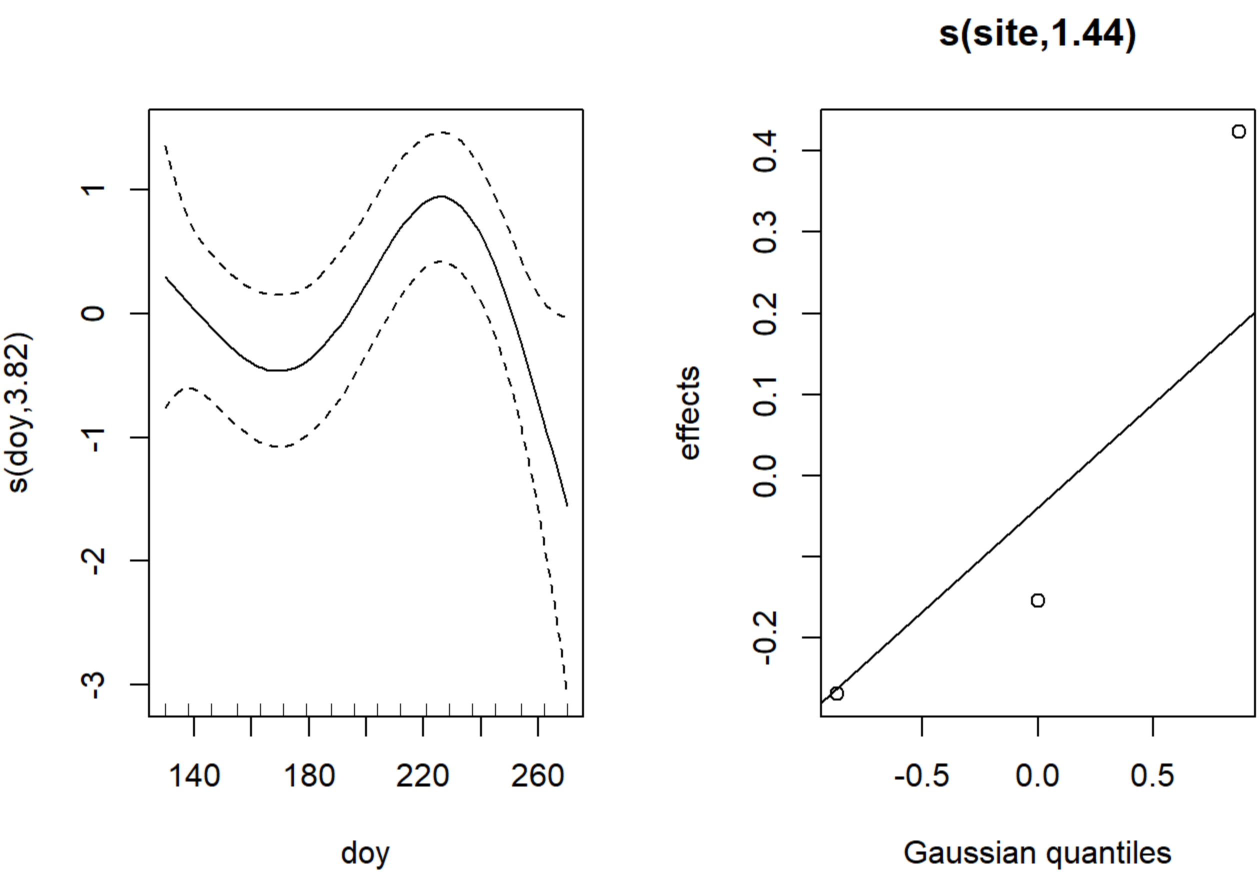
```
table(tick_dat$site)
```

```
##
##  A  B  C
## 36 36 36
```

```
m_gam <- gam(
  tick_count ~ s(doy, k = 8) + temp_capture + rh_capture + s(site, bs = "re"),
  family = nb(),
  data = tick_dat,
  method = "REML"
)

summary(m_gam)
```

```
##
## Family: Negative Binomial(1.534)
## Link function: log
##
## Formula:
## tick_count ~ s(doy, k = 8) + temp_capture + rh_capture + s(site,
##     bs = "re")
##
## Parametric coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.18362    1.23571  -1.767   0.0772 .
## temp_capture  0.17396    0.03460   5.028 4.95e-07 ***
## rh_capture   -0.03801    0.01555  -2.444   0.0145 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##           edf Ref.df Chi.sq p-value
## s(doy)  3.819  4.669  14.89  0.0089 **
## s(site) 1.441  2.000   5.02  0.0304 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.338   Deviance explained = 48.3%
## -REML = 112.21  Scale est. = 1         n = 108
```

```
plot(m_gam, pages = 1)
```



PART II — Ohio River Valley fish data (30 minutes) 8. Data description (tell students first)

This dataset contains fish survey observations (2010–2025) with:

-Common_Name

-Count

-Weight_kg

-River, County

-Latitude, Longitude

Each row = a species-specific observation at a place and time, not a population census.

9. Load the dataset

```
fish_dat <- read_excel(
  "2010-2025-Fish-Population-Data_for-web-1.xlsx"
)

glimpse(fish_dat)
```

```
## Rows: 55,328
## Columns: 33
## $ Event_Code       <chr> "013.7LDB072010", "013.7LDB072010", "013.7LDB072010",…
## $ LocCode          <chr> "OH013.7LDBb", "OH013.7LDBb", "OH013.7LDBb", "OH013.7…
## $ Comid            <chr> "3821213", "3821213", "3821213", "3821213", "3821213"…
## $ Latitude         <dbl> 40.55288, 40.55288, 40.55288, 40.55288, 40.55288, 40.…
## $ Longitude        <dbl> -80.21257, -80.21257, -80.21257, -80.21257, -80.21257…
## $ River            <chr> "Ohio River", "Ohio River", "Ohio River", "Ohio River…
## $ Rmi              <dbl> 13.7, 13.7, 13.7, 13.7, 13.7, 13.7, 13.7, 13.7, 13.7,…
## $ Bank             <chr> "LDB", "LDB", "LDB", "LDB", "LDB", "LDB", "LDB", "LDB…
## $ Pool             <chr> "Montgomery", "Montgomery", "Montgomery", "Montgomery…
## $ Ct               <chr> "EF", "EF", "EF", "EF", "EF", "EF", "EF", "EF", "EF",…
## $ Zonelength       <dbl> 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5…
## $ E_Time           <dbl> 2571, 2571, 2571, 2571, 2571, 2571, 2571, 2571, 2571,…
## $ Collector_Org    <chr> "ORSANCO", "ORSANCO", "ORSANCO", "ORSANCO", "ORSANCO"…
## $ Year             <dbl> 2010, 2010, 2010, 2010, 2010, 2010, 2010, 2010, 2010,…
## $ Date             <dttm> 2010-07-20, 2010-07-20, 2010-07-20, 2010-07-20, 2010…
## $ Temp_C           <dbl> 27.91, 27.91, 27.91, 27.91, 27.91, 27.91, 27.91, 27.9…
## $ Conductivity     <dbl> 445, 445, 445, 445, 445, 445, 445, 445, 445, 445, 445…
## $ Secchi_in        <dbl> 50, 50, 50, 50, 50, 50, 50, 50, 50, 50, 50, 50, 50, 5…
## $ State_LDB        <chr> "PA", "PA", "PA", "PA", "PA", "PA", "PA", "PA", "PA",…
## $ State_RDB        <chr> "PA", "PA", "PA", "PA", "PA", "PA", "PA", "PA", "PA",…
## $ County_LDB       <chr> "ALLEGHENY", "ALLEGHENY", "ALLEGHENY", "ALLEGHENY", "…
## $ County_RDB       <chr> "ALLEGHENY", "ALLEGHENY", "ALLEGHENY", "ALLEGHENY", "…
## $ USGS_Quad        <chr> "AMBRIDGE", "AMBRIDGE", "AMBRIDGE", "AMBRIDGE", "AMBR…
## $ PhyloNum         <dbl> 189, 232, 232, 221, 221, 221, 202, 191, 191, 189, 233…
## $ Common_Name      <chr> "Largemouth Bass", "Sauger", "Sauger", "Logperch", "L…
## $ Latin_Name       <chr> "Micropterus dolomieu", "Sander canadensis", "Sander …
## $ Count            <dbl> 5, 5, 7, 22, 1, 4, 1, 1, 1, 3, 1, 4, 3, 1, 5, 1, 3, 4…
## $ `3cm_Size_Class` <chr> "6.1 - 9.0", "27.1 - 30.0", "24.1 - 27.0", "9.1 - 12.…
## $ Length_cm        <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N…
## $ Weight_kg        <dbl> 0.038, 0.893, 0.877, 0.235, 0.005, 0.075, 0.003, 0.22…
## $ Weight_Est       <chr> "model 1-3-16", "model 1-3-16", "model 1-3-16", "mode…
## $ Voucher_Type     <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N…
## $ CatNo            <chr> "I-68543", "I-68549", "I-68549", "I-68547", "I-68547"…
```

```
head(fish_dat)
```

```
## # A tibble: 6 × 33
##   Event_Code     LocCode  Comid Latitude Longitude River   Rmi Bank  Pool  Ct
##   <chr>          <chr>    <chr>    <dbl>     <dbl> <chr> <dbl> <chr> <chr> <chr>
## 1 013.7LDB072010 OH013.7… 3821…     40.6     -80.2 Ohio…  13.7 LDB   Mont… EF
## 2 013.7LDB072010 OH013.7… 3821…     40.6     -80.2 Ohio…  13.7 LDB   Mont… EF
## 3 013.7LDB072010 OH013.7… 3821…     40.6     -80.2 Ohio…  13.7 LDB   Mont… EF
## 4 013.7LDB072010 OH013.7… 3821…     40.6     -80.2 Ohio…  13.7 LDB   Mont… EF
## 5 013.7LDB072010 OH013.7… 3821…     40.6     -80.2 Ohio…  13.7 LDB   Mont… EF
## 6 013.7LDB072010 OH013.7… 3821…     40.6     -80.2 Ohio…  13.7 LDB   Mont… EF
## # ℹ 23 more variables: Zonelength <dbl>, E_Time <dbl>, Collector_Org <chr>,
## #   Year <dbl>, Date <dttm>, Temp_C <dbl>, Conductivity <dbl>, Secchi_in <dbl>,
## #   State_LDB <chr>, State_RDB <chr>, County_LDB <chr>, County_RDB <chr>,
## #   USGS_Quad <chr>, PhyloNum <dbl>, Common_Name <chr>, Latin_Name <chr>,
## #   Count <dbl>, `3cm_Size_Class` <chr>, Length_cm <lgl>, Weight_kg <dbl>,
## #   Weight_Est <chr>, Voucher_Type <chr>, CatNo <chr>
```

```
tail(fish_dat)
```

```
## # A tibble: 6 × 33
##   Event_Code      LocCode Comid Latitude Longitude River   Rmi Bank  Pool  Ct
##   <chr>           <chr>   <chr>    <dbl>     <dbl> <chr> <dbl> <chr> <chr> <chr>
## 1 974.1LDBe080725 OH974.… <NA>      37.1     -89.2 Ohio…  974. LDB   Open… EF
## 2 974.1LDBe080725 OH974.… <NA>      37.1     -89.2 Ohio…  974. LDB   Open… EF
## 3 974.1LDBe080725 OH974.… <NA>      37.1     -89.2 Ohio…  974. LDB   Open… EF
## 4 974.1LDBe080725 OH974.… <NA>      37.1     -89.2 Ohio…  974. LDB   Open… EF
## 5 974.1LDBe080725 OH974.… <NA>      37.1     -89.2 Ohio…  974. LDB   Open… EF
## 6 974.1LDBe080725 OH974.… <NA>      37.1     -89.2 Ohio…  974. LDB   Open… EF
## # ℹ 23 more variables: Zonelength <dbl>, E_Time <dbl>, Collector_Org <chr>,
## #   Year <dbl>, Date <dttm>, Temp_C <dbl>, Conductivity <dbl>, Secchi_in <dbl>,
## #   State_LDB <chr>, State_RDB <chr>, County_LDB <chr>, County_RDB <chr>,
## #   USGS_Quad <chr>, PhyloNum <dbl>, Common_Name <chr>, Latin_Name <chr>,
## #   Count <dbl>, `3cm_Size_Class` <chr>, Length_cm <lgl>, Weight_kg <dbl>,
## #   Weight_Est <chr>, Voucher_Type <chr>, CatNo <chr>
```

```
summary(fish_dat)
```

```
##   Event_Code          LocCode            Comid              Latitude
##  Length:55328       Length:55328       Length:55328       Min.   :36.99
##  Class :character   Class :character   Class :character   1st Qu.:38.14
##  Mode  :character   Mode  :character   Mode  :character   Median :38.82
##                                                           Mean   :38.98
##                                                           3rd Qu.:39.84
##                                                           Max.   :40.70
##                                                           NA's   :1
##    Longitude          River                 Rmi              Bank
##  Min.   :-89.18   Length:55328       Min.   :  0.1   Length:55328
##  1st Qu.:-85.92   Class :character   1st Qu.:111.9   Class :character
##  Median :-82.88   Mode  :character   Median :344.6   Mode  :character
##  Mean   :-83.60                      Mean   :390.9
##  3rd Qu.:-80.82                      3rd Qu.:624.9
##  Max.   :-80.02                      Max.   :980.8
##  NA's   :1                           NA's   :556
##      Pool                Ct              Zonelength          E_Time
##  Length:55328       Length:55328       Min.   :   0.10   Min.   :    0
##  Class :character   Class :character   1st Qu.:   0.50   1st Qu.: 1932
##  Mode  :character   Mode  :character   Median :   0.50   Median : 2040
##                                        Mean   :   9.05   Mean   : 2121
##                                        3rd Qu.:   0.50   3rd Qu.: 2187
##                                        Max.   :4000.00   Max.   :21045
##                                        NA's   :858       NA's   :1497
##  Collector_Org           Year           Date
##  Length:55328       Min.   :2010   Min.   :2010-04-28 00:00:00.00
##  Class :character   1st Qu.:2012   1st Qu.:2012-10-04 00:00:00.00
##  Mode  :character   Median :2015   Median :2015-07-22 00:00:00.00
##                     Mean   :2016   Mean   :2016-05-16 09:37:05.26
##                     3rd Qu.:2018   3rd Qu.:2018-08-23 00:00:00.00
##                     Max.   :2025   Max.   :2025-08-20 00:00:00.00
##
##      Temp_C       Conductivity     Secchi_in       State_LDB
##  Min.   :17.50   Min.   :  0.0   Min.   :  6.00   Length:55328
##  1st Qu.:25.60   1st Qu.:339.0   1st Qu.: 24.00   Class :character
##  Median :27.40   Median :401.0   Median : 34.00   Mode  :character
##  Mean   :27.02   Mean   :395.3   Mean   : 37.03
##  3rd Qu.:28.92   3rd Qu.:465.0   3rd Qu.: 43.00
##  Max.   :34.74   Max.   :746.0   Max.   :118.00
##  NA's   :9606    NA's   :10520   NA's   :3519
##   State_RDB          County_LDB         County_RDB          USGS_Quad
##  Length:55328       Length:55328       Length:55328       Length:55328
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##     PhyloNum      Common_Name         Latin_Name            Count
##  Min.   :  2.0   Length:55328       Length:55328       Min.   :    0.000
##  1st Qu.: 78.0   Class :character   Class :character   1st Qu.:    1.000
```

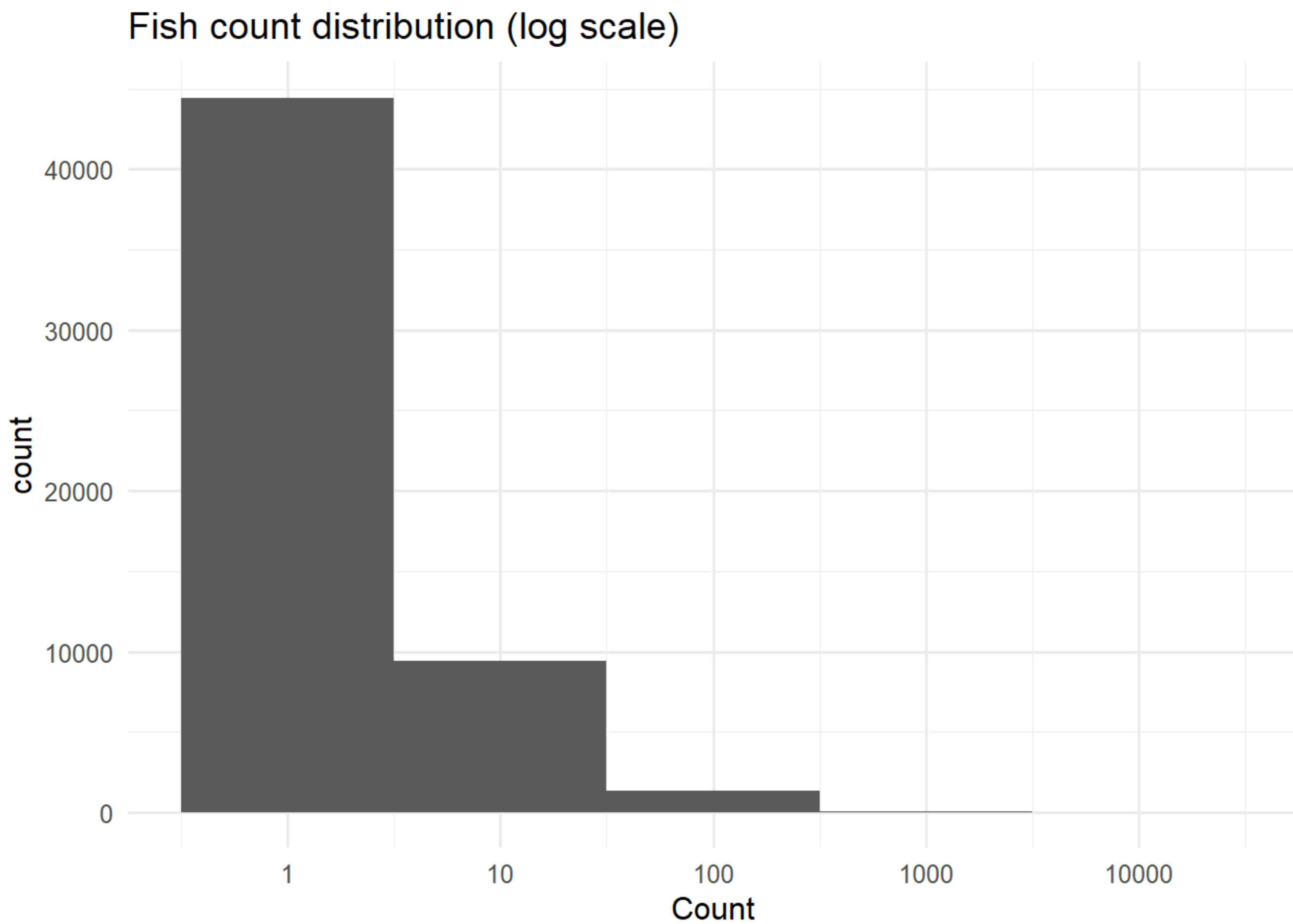
```
##  Median :125.0   Mode  :character   Mode  :character   Median :    1.000
##  Mean   :128.1                                         Mean   :    5.742
##  3rd Qu.:189.0                                         3rd Qu.:    3.000
##  Max.   :235.0                                         Max.   :12014.000
##                                                        NA's   :32
##  3cm_Size_Class     Length_cm          Weight_kg       Weight_Est
##  Length:55328       Mode :logical   Min.   : 0.000   Length:55328
##  Class :character   FALSE:3         1st Qu.: 0.030   Class :character
##  Mode  :character   TRUE :3972      Median : 0.178   Mode  :character
##                     NA's :51353     Mean   : 0.566
##                                     3rd Qu.: 0.614
##                                     Max.   :22.511
##                                     NA's   :18643
##  Voucher_Type          CatNo
##  Length:55328       Length:55328
##  Class :character   Class :character
##  Mode  :character   Mode  :character
##
##
##
##
```

10. Fingerprint #1 — counts by species

```
fish_dat %>%
  ggplot(aes(Count)) +
  geom_histogram(binwidth = 1) +
  scale_x_log10() +
  labs(title = "Fish count distribution (log scale)")
```

```
## Warning in scale_x_log10(): log-10 transformation introduced infinite values.
```

```
## Warning: Removed 33 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
names(fish_dat)
```

```
##  [1] "Event_Code"     "LocCode"        "Comid"          "Latitude"
##  [5] "Longitude"      "River"          "Rmi"            "Bank"
##  [9] "Pool"           "Ct"             "Zonelength"     "E_Time"
## [13] "Collector_Org"  "Year"           "Date"           "Temp_C"
## [17] "Conductivity"   "Secchi_in"      "State_LDB"      "State_RDB"
## [21] "County_LDB"     "County_RDB"     "USGS_Quad"      "PhyloNum"
## [25] "Common_Name"    "Latin_Name"     "Count"          "3cm_Size_Class"
## [29] "Length_cm"      "Weight_kg"      "Weight_Est"     "Voucher_Type"
## [33] "CatNo"
```

```
str(fish_dat$Count)
```

```
##  num [1:55328] 5 5 7 22 1 4 1 1 1 3 ...
```

```
summary(fish_dat$Count)
```

```
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.      NA's 
##     0.000     1.000     1.000     5.742     3.000 12014.000        32
```

```
#some nas that could be numbers treated like characters, lets try to force them to play nice
by parsing
fish_dat <- fish_dat %>%
  mutate(
    Count = parse_number(as.character(Count)),
    Weight_kg = parse_number(as.character(Weight_kg))
  )

mean(fish_dat$Count, na.rm = TRUE)
```

```
## [1] 5.741844
```

```
var(fish_dat$Count, na.rm = TRUE)
```

```
## [1] 3676.238
```

```
#the code below tells you proportionally how many zeros we have
n_zero <- sum(fish_dat$Count == 0, na.rm = TRUE)
n_obs  <- sum(!is.na(fish_dat$Count))

n_zero
```

```
## [1] 1
```

```
n_obs
```

```
## [1] 55296
```

```
n_zero / n_obs
```

```
## [1] 1.808449e-05
```

```
#only one zero row which is kinda odd for these data, let's take a look
fish_dat[fish_dat$Count == 0 & !is.na(fish_dat$Count), ]
```

```
## # A tibble: 1 × 33
##   Event_Code     LocCode  Comid Latitude Longitude River   Rmi Bank  Pool  Ct
##   <chr>          <chr>    <chr>    <dbl>     <dbl> <chr> <dbl> <chr> <chr> <chr>
## 1 041.0LDB062811 OH041.0… 3821…     40.6     -80.5 Ohio…    41 LDB   New … EF
## # ℹ 23 more variables: Zonelength <dbl>, E_Time <dbl>, Collector_Org <chr>,
## #   Year <dbl>, Date <dttm>, Temp_C <dbl>, Conductivity <dbl>, Secchi_in <dbl>,
## #   State_LDB <chr>, State_RDB <chr>, County_LDB <chr>, County_RDB <chr>,
## #   USGS_Quad <chr>, PhyloNum <dbl>, Common_Name <chr>, Latin_Name <chr>,
## #   Count <dbl>, `3cm_Size_Class` <chr>, Length_cm <lgl>, Weight_kg <dbl>,
## #   Weight_Est <chr>, Voucher_Type <chr>, CatNo <chr>
```

```
#that is likely an error so we can remove it here and have it documented as part of our data cleaning activity
fish_dat <- fish_dat %>%
  filter(is.na(Count) | Count != 0)
```

Do you notice any of the following?

-Extreme right skew

-Many zeros

-Mean ≪ variance

-Multiple species + locations

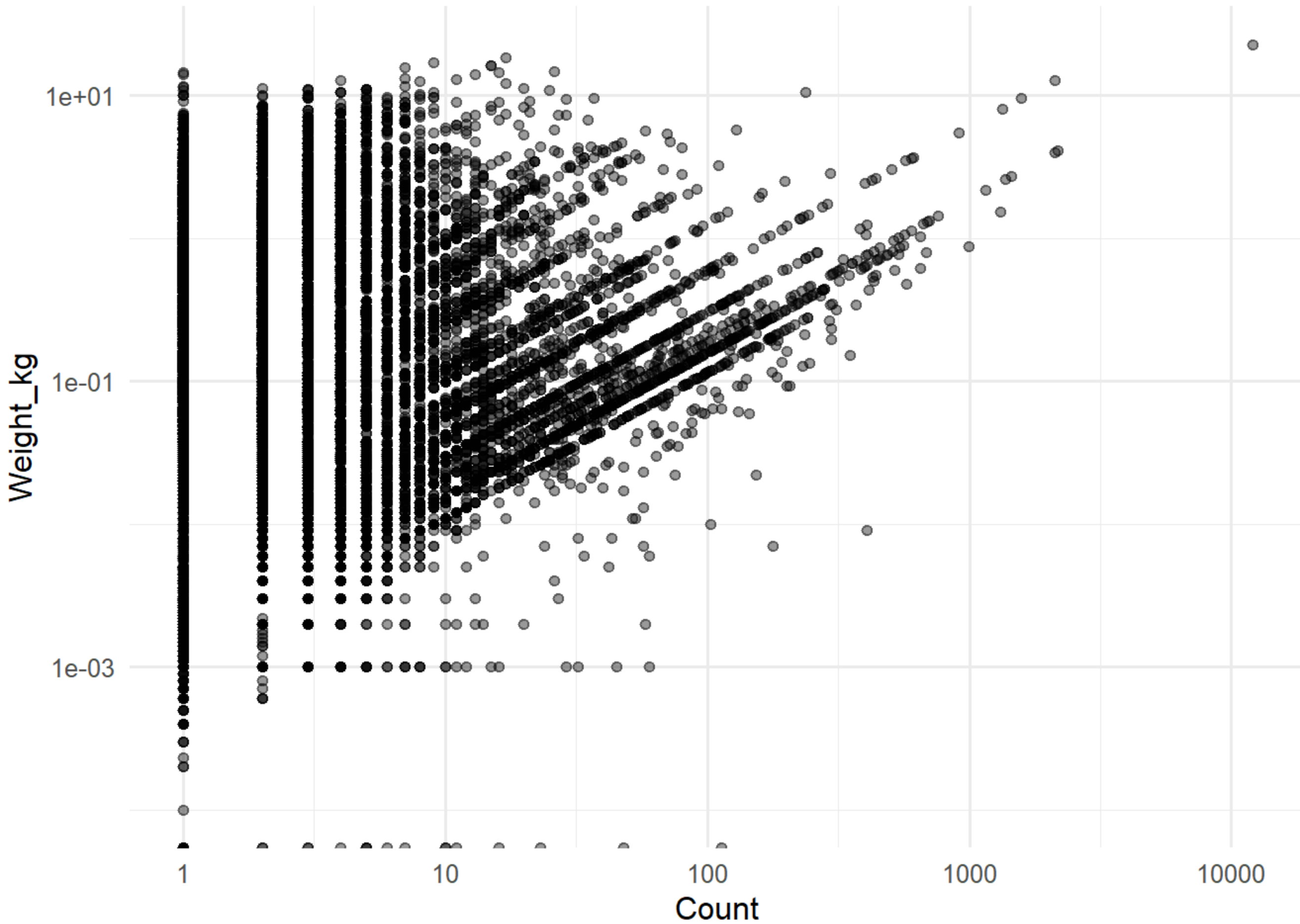
Immediate red flags: Poisson inappropriate, independence violated

11. Fingerprint #2 — weight vs count

```
fish_dat %>%
  ggplot(aes(Count, Weight_kg)) +
  geom_point(alpha = 0.4) +
  scale_x_log10() +
  scale_y_log10()
```

```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
```

```
## Warning: Removed 18643 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Interpretation

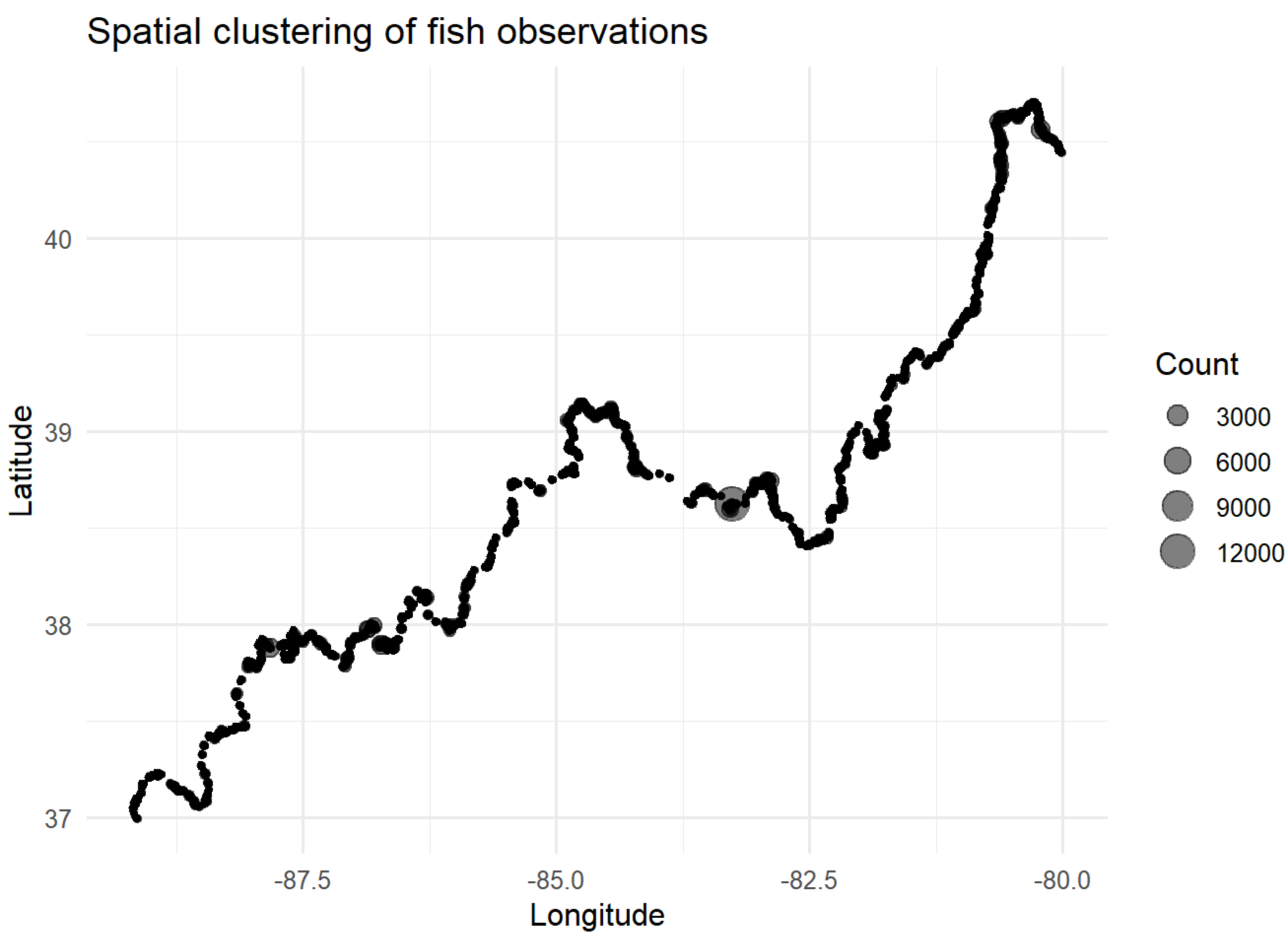
-Weight often log-normal

-Count and weight encode different ecological processes

-Suggests separate models or joint models 12. Fingerprint #3 — spatial structure

```
fish_dat %>%
  ggplot(aes(Longitude, Latitude, size = Count)) +
  geom_point(alpha = 0.5) +
  labs(title = "Spatial clustering of fish observations")
```

```
## Warning: Removed 33 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Spatial autocorrelation likely → Consider the following

-Mixed models

-Spatial smooths (GAMs plus spatial splines)

-Geostatistical approaches

13. Model sketching (not fitting everything!)

Example 1: Species-specific abundance

```
glmmTMB(
  Count ~ Common_Name + (1|River),
  family = nbinom2(),
  data = fish_dat
)
```

```
## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; function evaluation limit reached without convergence (9). See
## vignette('troubleshooting'), help('diagnose')
```

```
## Formula:          Count ~ Common_Name + (1 | River)
## Data: fish_dat
##       AIC       BIC    logLik -2*log(L)  df.resid
##  262959.9  264182.0 -131342.9  262685.9     55158
## Random-effects (co)variances:
##
## Conditional model:
##  Groups Name        Std.Dev.
##  River  (Intercept) 0.3472
##
## Number of obs: 55295 / Conditional model: River, 2
##
## Dispersion parameter for nbinom2 family (): 0.997
##
## Fixed Effects:
##
## Conditional model:
##                                (Intercept)
##                                  7.336e-01
##                     Common_NameBanded Darter
##                                  1.163e+00
##                  Common_NameBanded Killifish
##                                  7.473e-01
##                    Common_NameBanded Sculpin
##                                 -4.061e-01
##                       Common_NameBigeye Chub
##                                  3.627e-02
##                      Common_NameBighead Carp
##                                 -4.054e-01
##                  Common_NameBigmouth Buffalo
##                                 -2.284e-01
##                     Common_NameBlack Buffalo
##                                 -1.468e-01
##                    Common_NameBlack Bullhead
##                                 -4.061e-01
##                     Common_NameBlack Crappie
##                                 -2.838e-01
##                    Common_NameBlack Redhorse
##                                 -4.475e-02
##          Common_NameBlackstripe Topminnow
##                                  2.913e-01
##                      Common_NameBlue Catfish
##                                 -3.634e-01
##                       Common_NameBlue Sucker
##                                 -2.231e-01
##                 Common_NameBluebreast Darter
##                                  1.734e+00
##                         Common_NameBluegill
##                                  1.258e+00
##                  Common_NameBluntnose Minnow
##                                  1.271e+00
```

```
##                         Common_NameBowfin
##                                -1.177e-01
##                Common_NameBrook Silverside
##                                 7.040e-02
##                 Common_NameBullhead Minnow
##                                 5.185e-01
##                     Common_NameCarpiodes sp
##                                 4.417e-01
##            Common_NameCentral Stoneroller
##                                -1.600e-02
##                 Common_NameChannel Catfish
##                                 4.231e-01
##                  Common_NameChannel Darter
##                                 1.699e+00
##                  Common_NameChannel Shiner
##                                 3.038e+00
##           Common_NameChannel/Mimic Shiner
##                                 2.014e+00
##                Common_NameChestnut Lamprey
##                                -4.061e-01
##                     Common_NameCommon Carp
##                                -1.311e-01
##         Common_NameCommon Carp X Goldfish
##                                -4.061e-01
##                   Common_NameCommon Shiner
##                                 9.766e-01
##                      Common_NameCreek Chub
##                                -4.061e-01
##                    Common_NameDusky Darter
##                                 2.874e-01
##            Common_NameEastern Sand Darter
##                                 3.174e-06
##                  Common_NameEmerald Shiner
##                                 2.882e+00
##                  Common_NameFantail Darter
##                                -1.178e-01
##                Common_NameFlathead Catfish
##                                -2.982e-01
##                 Common_NameFreckled Madtom
##                                -4.061e-01
##                 Common_NameFreshwater Drum
##                                 1.391e+00
##                    Common_NameGhost Shiner
##                                 7.730e-01
##                     Common_NameGilt Darter
##                                -1.823e-01
##                    Common_NameGizzard Shad
##                                 2.580e+00
##                 Common_NameGolden Redhorse
##                                 4.503e-01
##                   Common_NameGolden Shiner
```

```
##                                  -4.054e-01
##                         Common_NameGoldeye
##                                  -4.064e-02
##                        Common_NameGoldfish
##                                  -4.053e-01
##                      Common_NameGrass Carp
##                                  -2.043e-01
##                  Common_NameGrass Pickerel
##                                  -4.061e-01
##                   Common_NameGreen Sunfish
##                                  -7.798e-02
##                 Common_NameGreenside Darter
##                                   2.395e-01
##               Common_NameHighfin Carpsucker
##                                   4.557e-02
##              Common_NameHybrid Striped Bass
##                                   6.427e-02
##                     Common_NameIctiobus sp
##                                  -4.061e-01
##               Common_NameInland Silverside
##                                   5.019e-01
##                   Common_NameJohnny Darter
##                                  -9.130e-02
##                 Common_NameLargemouth Bass
##                                   3.294e-01
##                  Common_NameLepomis Hybrid
##                                  -2.335e-01
##                      Common_NameLepomis sp
##                                   1.923e-01
##                  Common_NameLeuciscidae sp
##                                  -4.061e-01
##                        Common_NameLogperch
##                                   7.010e-01
##                 Common_NameLongear Sunfish
##                                   9.974e-01
##                 Common_NameLonghead Darter
##                                  -2.233e-01
##                   Common_NameLongnose Gar
##                                  -1.233e-01
##                  Common_NameMicropterus sp
##                                   5.746e-01
##         Common_NameMississippi Silverside
##                                  -4.042e-01
##     Common_NameMississippi Silvery Minnow
##                                   2.736e+00
##                         Common_NameMooneye
##                                  -2.250e-01
##                       Common_NameMorone Sp
##                                   1.094e+00
##                   Common_NameMoxostoma sp
##                                   4.418e-01
```

```
##                  Common_NameMud Darter
##                             -4.042e-01
##                Common_NameMuskellunge
##                             -1.432e-01
##           Common_NameNorthern Hogsucker
##                             -1.387e-01
##             Common_NameNorthern Madtom
##                             -4.061e-01
##               Common_NameNorthern Pike
##                             -4.054e-01
##                Common_NameNotropis sp
##                             -4.042e-01
##               Common_NameOhio Lamprey
##                             -4.051e-01
##        Common_NameOrangespotted Sunfish
##                              1.215e+00
##                 Common_NamePaddlefish
##                             -4.054e-01
##                 Common_NamePercina sp
##                             -4.061e-01
##               Common_NamePirate Perch
##                             -4.061e-01
##                Common_NamePumpkinseed
##                              3.323e-01
##                  Common_NameQuillback
##                              1.005e-02
##              Common_NameRainbow Darter
##                              8.766e-01
##              Common_NameRedear Sunfish
##                              3.932e-02
##               Common_NameRedfin Shiner
##                             -4.042e-01
##            Common_NameRiver Carpsucker
##                              4.067e-01
##                 Common_NameRiver Chub
##                             -4.051e-01
##               Common_NameRiver Darter
##                              1.086e+00
##             Common_NameRiver Redhorse
##                              1.116e-01
##               Common_NameRiver Shiner
##                              1.145e+00
##                  Common_NameRock Bass
##                              3.819e-01
##             Common_NameRosyface Shiner
##                             -4.053e-01
##                Common_NameSand Shiner
##                              2.130e+00
##                     Common_NameSauger
##                              4.345e-01
##                    Common_NameSaugeye
```

```
##                                  9.103e-01
##                       Common_NameShoal Chub
##                                  7.229e-01
##               Common_NameShorthead Redhorse
##                                 -1.955e-01
## Common_NameShorthead/Smallmouth Redhorse
##                                 -4.051e-01
##                    Common_NameShortnose Gar
##                                 -4.941e-03
##                      Common_NameSilver Carp
##                                  5.991e-02
##                      Common_NameSilver Chub
##                                  8.289e-01
##                   Common_NameSilver Lamprey
##                                 -4.054e-01
##                  Common_NameSilver Redhorse
##                                  1.070e-01
##                    Common_NameSilver Shiner
##                                  2.876e-01
##                 Common_NameSilverjaw Minnow
##                                 -4.042e-01
##                 Common_NameSkipjack Herring
##                                  1.984e-01
##               Common_NameSlenderhead Darter
##                                  4.788e-01
##               Common_NameSmallmouth Buffalo
##                                  1.042e-01
##              Common_NameSmallmouth Redhorse
##                                  2.978e-01
##                   Common_NameSpotfin Shiner
##                                  9.807e-01
##                  Common_NameSpottail Shiner
##                                  5.085e-01
##                    Common_NameSpotted Bass
##                                  3.010e-01
##                     Common_NameSpotted Gar
##                                 -3.383e-01
##                  Common_NameSpotted Sucker
##                                  2.766e-02
##                Common_NameSteelcolor Shiner
##                                  6.904e-01
##                        Common_NameStonecat
##                                 -4.061e-01
##                  Common_NameStreamline Chub
##                                  4.569e-01
##                     Common_NameStriped Bass
##                                  6.826e-04
##                   Common_NameStriped Mullet
##                                  1.329e+00
##                   Common_NameStriped Shiner
##                                  1.428e-01
```

```
##            Common_NameSuckermouth Minnow
##                               -1.177e-01
##                Common_NameThreadfin Shad
##                                1.494e+00
##             Common_NameTippecanoe Darter
##                                1.003e+00
##                   Common_NameTrout-Perch
##                                8.724e-01
##              Common_NameVariegate Darter
##                               -4.061e-01
##                       Common_NameWalleye
##                               -1.211e-03
##                     Common_NameWarmouth
##                               -2.727e-02
##        Common_NameWestern Blacknose Dace
##                                2.913e-01
##          Common_NameWestern Mosquitofish
##                                1.974e+00
##                    Common_NameWhite Bass
##                                2.340e-01
##                 Common_NameWhite Crappie
##                               -2.379e-01
##                   Common_NameWhite Perch
##                               -2.513e-01
##                  Common_NameWhite Sucker
##                               -4.053e-01
##                   Common_NameYellow Bass
##                                1.557e-02
##              Common_NameYellow Bullhead
##                               -4.051e-01
##                  Common_NameYellow Perch
##                                1.650e-01
```

Example 2: Zero-inflated detection model

```
glmmTMB(
  Count ~ Common_Name + (1|River),
  ziformula = ~1,
  family = nbinom2(),
  data = fish_dat
)
```

```
## Formula:          Count ~ Common_Name + (1 | River)
## Zero inflation:         ~1
## Data: fish_dat
##       AIC       BIC    logLik -2*log(L)  df.resid
##  262961.9  264192.9 -131342.9  262685.9     55157
## Random-effects (co)variances:
##
## Conditional model:
##  Groups Name        Std.Dev.
##  River  (Intercept) 0.3474
##
## Number of obs: 55295 / Conditional model: River, 2
##
## Dispersion parameter for nbinom2 family (): 0.997
##
## Fixed Effects:
##
## Conditional model:
##                             (Intercept)
##                               0.7332273
##                  Common_NameBanded Darter
##                               1.1623555
##               Common_NameBanded Killifish
##                               0.7476816
##                 Common_NameBanded Sculpin
##                              -0.4050938
##                    Common_NameBigeye Chub
##                               0.0367999
##                   Common_NameBighead Carp
##                              -0.4050377
##               Common_NameBigmouth Buffalo
##                              -0.2273543
##                  Common_NameBlack Buffalo
##                              -0.1462205
##                 Common_NameBlack Bullhead
##                              -0.4050938
##                  Common_NameBlack Crappie
##                              -0.2833118
##                 Common_NameBlack Redhorse
##                              -0.0440168
##          Common_NameBlackstripe Topminnow
##                               0.2881982
##                   Common_NameBlue Catfish
##                              -0.3633659
##                    Common_NameBlue Sucker
##                              -0.2227135
##              Common_NameBluebreast Darter
##                               1.7350696
##                      Common_NameBluegill
##                               1.2589792
##               Common_NameBluntnose Minnow
```

```
##                                  1.2711375
##                          Common_NameBowfin
##                                 -0.1173532
##                Common_NameBrook Silverside
##                                  0.0715190
##                  Common_NameBullhead Minnow
##                                  0.5187542
##                      Common_NameCarpiodes sp
##                                  0.4423038
##               Common_NameCentral Stoneroller
##                                 -0.0150118
##                  Common_NameChannel Catfish
##                                  0.4237327
##                   Common_NameChannel Darter
##                                  1.6991387
##                   Common_NameChannel Shiner
##                                  3.0387760
##             Common_NameChannel/Mimic Shiner
##                                  2.0153694
##                 Common_NameChestnut Lamprey
##                                 -0.4050938
##                     Common_NameCommon Carp
##                                 -0.1303776
##          Common_NameCommon Carp X Goldfish
##                                 -0.4050938
##                   Common_NameCommon Shiner
##                                  0.9812749
##                      Common_NameCreek Chub
##                                 -0.4050938
##                    Common_NameDusky Darter
##                                  0.2881605
##              Common_NameEastern Sand Darter
##                                  0.0004364
##                  Common_NameEmerald Shiner
##                                  2.8824972
##                  Common_NameFantail Darter
##                                 -0.1173494
##                Common_NameFlathead Catfish
##                                 -0.2975374
##                 Common_NameFreckled Madtom
##                                 -0.4050938
##                 Common_NameFreshwater Drum
##                                  1.3914694
##                    Common_NameGhost Shiner
##                                  0.7736434
##                     Common_NameGilt Darter
##                                 -0.1818915
##                   Common_NameGizzard Shad
##                                  2.5801731
##                 Common_NameGolden Redhorse
##                                  0.4509366
```

```
##                    Common_NameGolden Shiner
##                                  -0.4050434
##                          Common_NameGoldeye
##                                  -0.0404138
##                         Common_NameGoldfish
##                                  -0.4050447
##                       Common_NameGrass Carp
##                                  -0.2043500
##                   Common_NameGrass Pickerel
##                                  -0.4050938
##                    Common_NameGreen Sunfish
##                                  -0.0772757
##                 Common_NameGreenside Darter
##                                   0.2400765
##               Common_NameHighfin Carpsucker
##                                   0.0461371
##                Common_NameHybrid Striped Bass
##                                   0.0646901
##                     Common_NameIctiobus sp
##                                  -0.4050938
##                Common_NameInland Silverside
##                                   0.5026226
##                    Common_NameJohnny Darter
##                                  -0.0913893
##                  Common_NameLargemouth Bass
##                                   0.3300328
##                   Common_NameLepomis Hybrid
##                                  -0.2332515
##                       Common_NameLepomis sp
##                                   0.1927511
##                   Common_NameLeuciscidae sp
##                                  -0.4050938
##                         Common_NameLogperch
##                                   0.7014592
##                  Common_NameLongear Sunfish
##                                   0.9981054
##                  Common_NameLonghead Darter
##                                  -0.2227054
##                     Common_NameLongnose Gar
##                                  -0.1226713
##                   Common_NameMicropterus sp
##                                   0.5751777
##           Common_NameMississippi Silverside
##                                  -0.4050319
##       Common_NameMississippi Silvery Minnow
##                                   2.7361279
##                          Common_NameMooneye
##                                  -0.2244797
##                        Common_NameMorone Sp
##                                   1.0941612
##                     Common_NameMoxostoma sp
```

```
##                                       0.4423445
##                            Common_NameMud Darter
##                                      -0.4050319
##                           Common_NameMuskellunge
##                                      -0.1426791
##                   Common_NameNorthern Hogsucker
##                                      -0.1379696
##                      Common_NameNorthern Madtom
##                                      -0.4050938
##                        Common_NameNorthern Pike
##                                      -0.4050434
##                          Common_NameNotropis sp
##                                      -0.4050319
##                         Common_NameOhio Lamprey
##                                      -0.4050398
##                Common_NameOrangespotted Sunfish
##                                       1.2154615
##                           Common_NamePaddlefish
##                                      -0.4050377
##                           Common_NamePercina sp
##                                      -0.4050938
##                         Common_NamePirate Perch
##                                      -0.4050938
##                          Common_NamePumpkinseed
##                                       0.3329526
##                            Common_NameQuillback
##                                       0.0106839
##                       Common_NameRainbow Darter
##                                       0.8777419
##                       Common_NameRedear Sunfish
##                                       0.0401595
##                        Common_NameRedfin Shiner
##                                      -0.4050319
##                     Common_NameRiver Carpsucker
##                                       0.4074105
##                           Common_NameRiver Chub
##                                      -0.4050398
##                         Common_NameRiver Darter
##                                       1.0871696
##                       Common_NameRiver Redhorse
##                                       0.1124611
##                         Common_NameRiver Shiner
##                                       1.1454862
##                            Common_NameRock Bass
##                                       0.3825946
##                      Common_NameRosyface Shiner
##                                      -0.4050447
##                          Common_NameSand Shiner
##                                       2.1299797
##                               Common_NameSauger
##                                       0.4351676
```

```
##                          Common_NameSaugeye
##                                   0.9108194
##                       Common_NameShoal Chub
##                                   0.7234420
##               Common_NameShorthead Redhorse
##                                  -0.1952816
## Common_NameShorthead/Smallmouth Redhorse
##                                  -0.4050398
##                    Common_NameShortnose Gar
##                                  -0.0043504
##                      Common_NameSilver Carp
##                                   0.0604306
##                      Common_NameSilver Chub
##                                   0.8294214
##                   Common_NameSilver Lamprey
##                                  -0.4050377
##                  Common_NameSilver Redhorse
##                                   0.1076705
##                    Common_NameSilver Shiner
##                                   0.2880832
##                 Common_NameSilverjaw Minnow
##                                  -0.4050319
##                 Common_NameSkipjack Herring
##                                   0.1991495
##               Common_NameSlenderhead Darter
##                                   0.4790306
##               Common_NameSmallmouth Buffalo
##                                   0.1048943
##              Common_NameSmallmouth Redhorse
##                                   0.2983884
##                   Common_NameSpotfin Shiner
##                                   0.9813726
##                  Common_NameSpottail Shiner
##                                   0.5092704
##                    Common_NameSpotted Bass
##                                   0.3016598
##                     Common_NameSpotted Gar
##                                  -0.3384485
##                  Common_NameSpotted Sucker
##                                   0.0278359
##               Common_NameSteelcolor Shiner
##                                   0.6935858
##                       Common_NameStonecat
##                                  -0.4050938
##                 Common_NameStreamline Chub
##                                   0.4561497
##                    Common_NameStriped Bass
##                                   0.0004100
##                  Common_NameStriped Mullet
##                                   1.3295985
##                  Common_NameStriped Shiner
```

```
##                                  0.1435336
##              Common_NameSuckermouth Minnow
##                                 -0.1173532
##                  Common_NameThreadfin Shad
##                                  1.4950396
##               Common_NameTippecanoe Darter
##                                  1.0037501
##                     Common_NameTrout-Perch
##                                  0.8728608
##                Common_NameVariegate Darter
##                                 -0.4050938
##                        Common_NameWalleye
##                                 -0.0006839
##                       Common_NameWarmouth
##                                 -0.0267434
##          Common_NameWestern Blacknose Dace
##                                  0.2881982
##            Common_NameWestern Mosquitofish
##                                  1.9745613
##                      Common_NameWhite Bass
##                                  0.2346669
##                   Common_NameWhite Crappie
##                                 -0.2374029
##                     Common_NameWhite Perch
##                                 -0.2508884
##                    Common_NameWhite Sucker
##                                 -0.4050447
##                     Common_NameYellow Bass
##                                  0.0161760
##                 Common_NameYellow Bullhead
##                                 -0.4050398
##                    Common_NameYellow Perch
##                                  0.1659140
##
## Zero-inflation model:
## (Intercept)
##       -22.1
```

Data fingerprint	What it tells you	Plausible tools
Mean ≪ variance	Overdispersion	Negative binomial
Many zeros	Detection / availability	Zero-inflated models
Repeated sites	Non-independence	Mixed models
Skewed weights	Multiplicative growth	Log-normal models
Spatial clustering	Shared environment	Spatial random effects
Extreme covariates dominate	Model chasing boundaries	Caution with MaxEnt